A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Impacts of agriculture, social/political factors & sustainable agriculture - Set 1

Mark scheme

AQA spec 3.5.1.3-3.5.1.5; Biological resources; textbook pp. 328-337

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2** Drainage makes soils aerobic and drier, increasing grass growth but wetland plants and the animals they support (waders, amphibians, invertebrates) cannot survive **2** Fertiliser increases nutrient availability so vigorous species (grasses, nettles) grow taller and out-compete smaller wildflowers for light – diversity falls **2** Hedgerow removal destroys habitat / corridors and nesting sites – loss of natural pest predator habitats increases pest damage – larger fields increase wind erosion – genetic contamination and introduced species also accepted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Long-continued traditional farming has created plagioclimax habitats colonised by many species: hay meadows, chalk grassland, grazed heath / moorland, hedgerows **1** These communities depend on the farming continuing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **plagioclimax** → a community held back from its climax by human activity	2
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** CO2 from fossil fuel use in machinery and embodied energy of fertiliser (Haber process) and machinery – ploughing increases soil respiration **1** Methane from livestock intestines and rice paddies – nitrous oxide from manure and nitrogen fertilisers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-330; AQA spec 3.5.1.3.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** Irrigation depletes rivers and aquifers (eg Aral Sea, Ogallala) – increases evaporation **1** Soil erosion and compaction reduce water retention and infiltration so runoff and river flow fluctuations increase **1 Evapotranspiration changes:** increased in irrigated arid areas, decreased where forests were cleared **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infiltration** → water soaking into the soil surface • **Evapotranspiration** → evaporation plus transpiration **Do not credit** • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	3

Do not write outside the box

Question	Marking guidance	Marks
02.2	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Seasonal / local:** avoids energy for heating, lighting greenhouses out of season and long-distance transport (food miles) and the pollution this causes **2 Organic:** uses natural processes (rotation, legumes, manure, biological control, predator habitats) instead of artificial fertilisers and pesticides, so less pollution, energy use and harm to non-target species – free-range and Fairtrade also accepted for welfare / social reasons **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
02.3	**1 mark each** 1 mark for each of the 3 points below. **1** Food shortages after 1945 (dependence on US food aid) – farmers could not afford loans to invest so grants funded machinery, drainage, hedgerow removal, liming, improved livestock **1 Guaranteed market / price support:** government bought surpluses to hold prices up (and sold stores in poor years) giving farmers security to increase output **1 Consequences:** loss of hedgerows and wetlands, intensification, pesticide and fertiliser pollution – later food surpluses (butter mountains) that could not be sold without harming LEDC producers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-333; AQA spec 3.5.1.3-3.5.1.4.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Other MEDCs (USA, Canada, Australia) also had surpluses – Japan already had suppliers **1** Eastern Europe / USSR needed food but could not pay **1** Selling cheaply in LEDCs would undercut local farmers, put them out of business and reduce their long-term food production **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 4 points below. **1** Quotas limiting production (eg milk) **1** Set-aside payments for taking land out of production **1** Farm diversification (recreation, farm shops, specialised products) and alternative crops / livestock (biofuels, pharmaceutical poppies, llamas, deer, bees) **1** Agri-environmental schemes (ESA, CSS, ESS, CS) paying for environmentally beneficial farming **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
03.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Hedgerows at least 1.5 m tall cut at 2-year intervals outside the breeding season - bird / insect habitat and landscape **1** Buffer strips / field margins (no cultivation, fertiliser or pesticide within 2 m) - protect ditches and rivers, habitat **1** Beetle banks of tussocky grass - natural pest predators **1** Skylark nesting plots – wild bird seed / nectar plants - winter food and pollinators **1** In-field tree protection – controlled stocking density to reduce compaction – ditch management – dry-stone walls – woodland management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 332-334; AQA spec 3.5.1.4.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Why needed:** growing population (to 12 billion), land degradation since the fertile crescent, reliance on artificial fertilisers, pesticides and unsustainable irrigation – limits to new land and imports • **Pest control:** cultural methods (weeding, mulching, rotation, barrier / companion crops, predator habitats, biological control, sterile males, pheromones), resistant varieties, integrated control, reduced antibiotics • **Nutrients:** natural processes - nitrogen-fixing legumes, decomposition, recycling organic matter (manure, compost), crop rotation, permaculture, conservation of soil biota and mycorrhizae, reduced artificial fertiliser – phosphate recycling • **Energy:** natural processes instead of Haber-process fertiliser, low tillage, less machinery, renewable energy, low food miles, reduced methane (cattle diets, rice varieties) • **Water:** low water-use crops, drip irrigation, reservoirs and aquifer recharge, avoiding saline water – soil: organic matter maintenance, erosion control • **Biodiversity:** retaining hedgerows, ditches, ponds, woodlands – CWR conservation and seed banks – agri-environment schemes • **Social:** control of impacts on rural communities, Fairtrade, local food – consumer choices • **Evaluation:** yields vs sustainability, cost, need for policy support – judgement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9

Do not write outside the box

Question	Marking guidance	Marks
	• **sustainability** → meeting present needs without stopping future generations meeting theirs • **biological control** → using a natural predator, parasite or pathogen of the pest • **nitrogen-fixing** → nitrogen gas converted into ammonium or other nitrogen compounds **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	
	Model answer - Level 3 (8 marks)	
	Sustainability matters because the population may reach 12 billion, land has been degraded since farming began in the fertile crescent, and current yields depend on artificial fertilisers, pesticides and irrigation that draws down groundwater, with little new land to cultivate. Pest control can move from chemicals to cultural methods - weeding, mulching, rotation, barrier and companion crops, habitats for predators - biological control, sterile males and pheromones, resistant varieties and integrated control, with fewer antibiotics in livestock.	
	Nutrients can come from natural processes: legumes fixing nitrogen, recycling manure and compost, rotation, permaculture and conserving soil biota and mycorrhizae, reducing manufactured nitrate and the non-renewable rock phosphate it accompanies. Energy use falls with low tillage, less machinery, renewable power, low food miles and lower methane from cattle diets and drought-tolerant rice. Water is conserved with low water-use crops, drip irrigation, reservoirs and aquifer recharge and by avoiding saline water; soil by maintaining organic matter and controlling erosion. Retaining hedgerows, ditches, ponds and woodland, conserving crop wild relatives in seed banks and agri-environment schemes protect the biodiversity that supplies pollination and pest control, while Fairtrade, local food and consumer choices address the social side.	
	The tension is between yield and sustainability: some strategies cost more or yield less in the short term and need policy support to be adopted. But since the unsustainable inputs themselves depend on finite fossil fuels and phosphate, the strategies that use natural processes are the only ones that can feed a larger population indefinitely.	
	Weaker answer - Level 2 (4 marks)	
	Agriculture can be made more sustainable by using less pesticides and using biological control instead, using manure instead of artificial fertilisers, growing legumes which add nitrogen to the soil, and using crop rotation. Drip irrigation saves water. Farmers can keep hedgerows to provide habitats for wildlife. Organic farming uses these methods. These strategies reduce pollution and keep the soil healthy so food can be produced in the future.	
	What makes the difference	
	• Level 2 lists valid strategies but does not explain why they are needed (population growth, land degradation, dependence on finite inputs) or group them by the problem they solve.	
	• Level 3 covers energy, soil, biodiversity and social strategies as well as pests, nutrients and water.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3 discusses the yield-sustainability tension and the need for policy support; Level 2 has no evaluation.	
04.2	1 mark for the point below. **1** Agricultural systems designed to mimic natural ecosystems - perennial crops, mixed species, recycling of nutrients, minimal external inputs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	
05.1	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Hedgerows **1** Hay meadows **1** Grazed moorland / heathland **1** Chalk grassland **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive **1** Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
05.3	1 mark for the point below. **1** The introduced (non-indigenous) predator may eat other species as well as the pest **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** Surveys monitor photosynthesis rates, biomass, soil water content and the spread of pests / diseases so inputs are applied only where needed **1** Yield mapping of different parts of a field controls fertiliser application in each area for future crops **1** This maximises yield while avoiding wastage / reducing leaching and nutrient pollution **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leaching** → nutrients or ions washed down out of the soil / washed away by water	3
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Machinery, pesticides, fertilisers, improved breeds and infrastructure (transport, refrigeration) are expensive **1** They are mainly available in affluent societies – poorer farmers cannot afford the inputs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2
06.3	1 mark for the point below. **1** Provides an income that allows producers to afford basic rights such as water, education, health care and food **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Nitrates are leached into water bodies used as sources of potable water **1** High nitrate levels cause blue baby syndrome (methaemoglobinaemia) and nitrates may be carcinogenic **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water	2
06.5	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Anaerobic digestion by microbes in rice padi fields **1** Livestock intestines (ruminant digestion) **1** Anaerobic decay of manure / slurry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-332; AQA spec 3.5.1.3-3.5.1.4.	
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** About 10% of land is already cultivated and 25% grazed **1** Most remaining land is too dry, wet, cold, hot or steep, is covered in ice, or has no soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Land becomes too expensive for local people to rent / is used to grow export crops for more affluent societies **1** So people with less money lose access to the land and food they used to produce **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** It uses natural processes to supply nutrients (legumes, recycled organic matter, soil biota), control pests and provide ecological services **1** So that pesticides and artificial fertilisers become unnecessary rather than just being banned **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2

Do not write outside the box

Question	Marking guidance	Marks
07.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Unsustainable features:** reliance on chemical pesticides (resistance, non-target impacts, bans), non-renewable rock phosphate, fossil-fuel-intensive nitrate manufacture, over-exploitation of rivers / groundwater, salinisation, loss of soil organic matter, greenhouse gases from fuel, rice and livestock, loss of species providing ecosystem services • **Pest control:** cultural methods (weeding, mulching, crop rotation, barrier crops, culling), biological control, predator habitats, polyculture / companion crops, integrated control, reduced antibiotic use • **Nutrients:** recycling organic matter, crop rotation, cultivation of legumes, conservation of soil biota, reduced artificial fertiliser especially nitrate, low tillage • **Water:** low water-use crops, reservoirs and aquifer recharge, drip irrigation rather than sprays • **Soil and carbon:** low-tillage farming to maintain organic matter, reduced machinery, renewable energy, drought-tolerant rice varieties so fields are drained earlier, high-carbohydrate ground feed for cattle reduces methane • **Biodiversity:** conserve habitats of crop wild relatives, seed banks, retain hedgerows, ditches, ponds, woodland – agri-environment schemes • **Evaluation:** growing population (perhaps 12 billion) needs more food – lower trophic level diets – obstacles - cost, need for support, gaining acceptance of new techniques – learning from past land degradation (fertile crescent, US mid-west) • **Judgement:** no single strategy sufficient – combination of ecological principles and new technology – justified conclusion **Also accept**	9

Do not write outside the box

Question	Marking guidance	Marks
	• Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **salinisation** → salt building up in the soil / soil becoming salty • **biological control** → using a natural predator, parasite or pathogen of the pest **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	
	Model answer - Level 3 (8 marks)	
	Present agriculture relies on inputs that cannot last: chemical pesticides to which pests evolve resistance and which harm non-target species, rock phosphate that is non-renewable, nitrate fertiliser made with fossil-fuel energy, and irrigation that over-draws rivers and groundwater and salinises soil, while ploughing loses soil organic matter and livestock, rice and machinery release greenhouse gases. Each can be replaced by strategies based on ecological principles. Pests are controlled culturally by weeding, mulching, rotation, barrier crops and culling, by biological control and predator habitats, by polyculture and companion crops, and by integrated control that uses chemicals only as a last resort. Nutrients come from recycled organic matter, rotation with legumes and conserved soil biota, so artificial fertiliser, especially nitrate, can be reduced. Water is saved with low water-use crops, reservoirs and aquifer recharge and drip irrigation; soil carbon is kept by low tillage and less machinery, with renewable energy for what remains; methane falls with rice varieties that allow earlier drainage and with ground, carbohydrate-rich cattle feed. Habitats of crop wild relatives, seed banks, hedgerows, ditches, ponds and woodland and agri-environment schemes protect the biodiversity that supplies these services.	
	Food demand will rise with a population that may reach 12 billion, there is little new land, and imports can reduce food availability where it is grown, so yield cannot be ignored. Some strategies cost more or need farmers to accept new techniques, and past land degradation from the fertile crescent to the American mid-west shows what happens when they are not adopted. No single strategy suffices; a combination of ecological principles with new technology is the only route to feeding more people without exhausting the resources food depends on.	
	Weaker answer - Level 1 (2 marks)	
	To make agriculture sustainable farmers should use fewer chemicals. Organic farming does not use pesticides or artificial fertilisers. Crop rotation and manure keep the soil fertile. Farmers should not waste water. Hedges should be kept for wildlife. This would make farming more sustainable but it might produce less food.	
	What makes the difference	
	• Level 1 gives general advice; Level 3 identifies each unsustainable feature and matches it with specific strategies, showing understanding of why they work.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3's evaluation considers population growth, land limits, cost, adoption and historical evidence; Level 1's is a single clause.	
	• Note that Level 3 avoids defining organic farming as 'not using chemicals' - the textbook stresses it is the positive use of natural processes.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	

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